

Year 9 Science Biological Sciences Heart Dissection

INVESTIGATION 4.7

Heart dissection

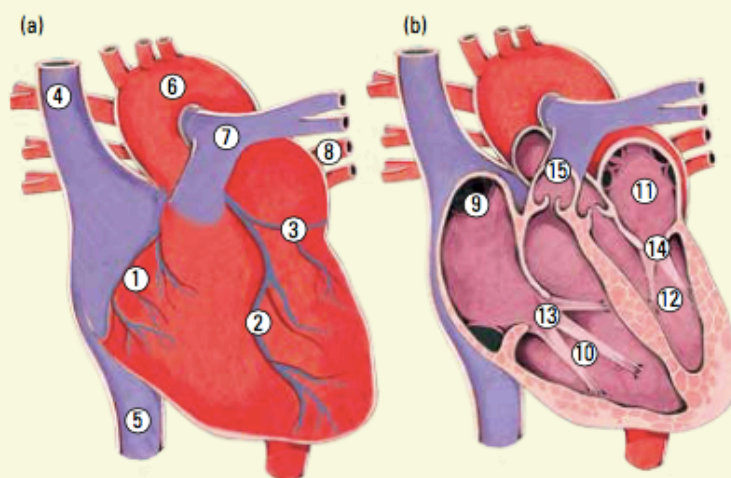
AIM To observe the structure of a mammalian heart

Materials:

sheep's heart preferably with the blood vessels still attached
dissecting instruments
dissecting board
newspaper or paper to cover dissection board

METHOD AND RESULTS

- ▶ Place newspaper on the dissection board, then place the heart on top of the paper.
- ▶ Use the diagram on the right to identify the parts of the heart.
- ▶ Try to locate where blood enters and leaves the heart:
 - (a) to and from the lungs
 - (b) to and from the rest of the body.
- 1** Sketch and label the heart and use arrows to show the direction of blood flow.
- ▶ Cut the heart in two so that both halves show the two sides of the heart (similar to the illustration on the previous page).
- 2** In a diagram, record your observations of the thickness of the walls on the left side of the heart compared with the right side.



- | | |
|---|---------------------|
| 1. Right coronary artery | 8. Pulmonary vein |
| 2. Left anterior descending coronary artery | 9. Right atrium |
| 3. Circumflex coronary artery | 10. Right ventricle |
| 4. Superior vena cava | 11. Left atrium |
| 5. Inferior vena cava | 12. Left ventricle |
| 6. Aorta | 13. Tricuspid valve |
| 7. Pulmonary artery | 14. Mitral valve |
| | 15. Pulmonary valve |

- 3** Suggest reasons for the differences observed.
- ▶ Try to locate the valves in the heart.

DISCUSS AND EXPLAIN

- 4** Describe the valves and suggest their function.
- 5** Write a summary paragraph about the structure and function of the heart.